Primer for Knots and Links Reading Course

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September 10th, 2020

Knots? Links?

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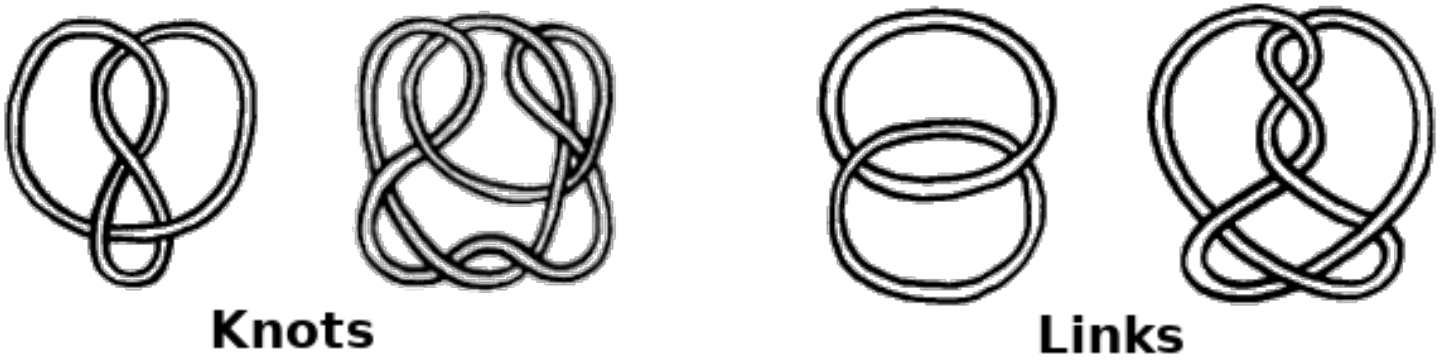

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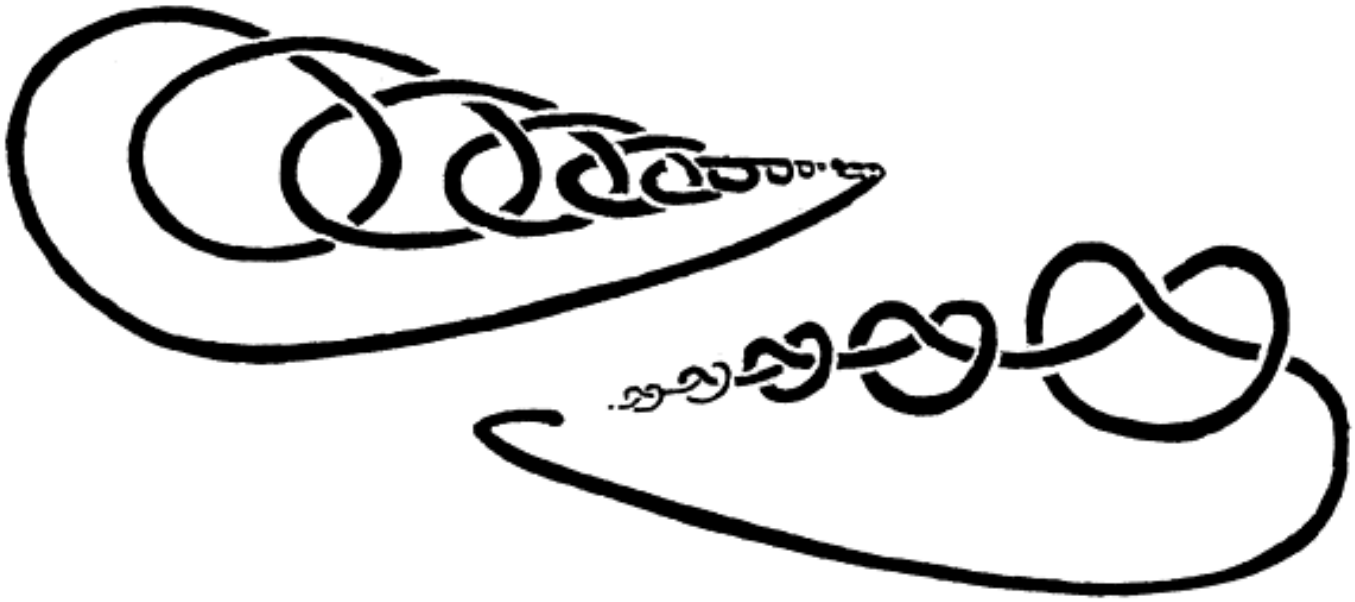
Definition (Tame knot)

We say that a knot $f : S^1 \to S^3$ is **tame** (polygonal) if it's image is ambient isotopic to a finite union of line segments. Otherwise, we say that f is **wild**. Some examples of the latter are pictured below.



The fundamental question

We say that two knots $K_1, K_2 \subset S^n$ are of the same **knot type** if they are ambient isotopic, i.e. there is a continuously varying collection of homeomorphisms

$$H_t : S^n \to S^n \qquad H_0 = Id_{S^n},\ H_1|_{K_1} = K_2$$

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So really, when we're talking about knots, we're talking about an equivalence class of embeddings under ambient isotopy.

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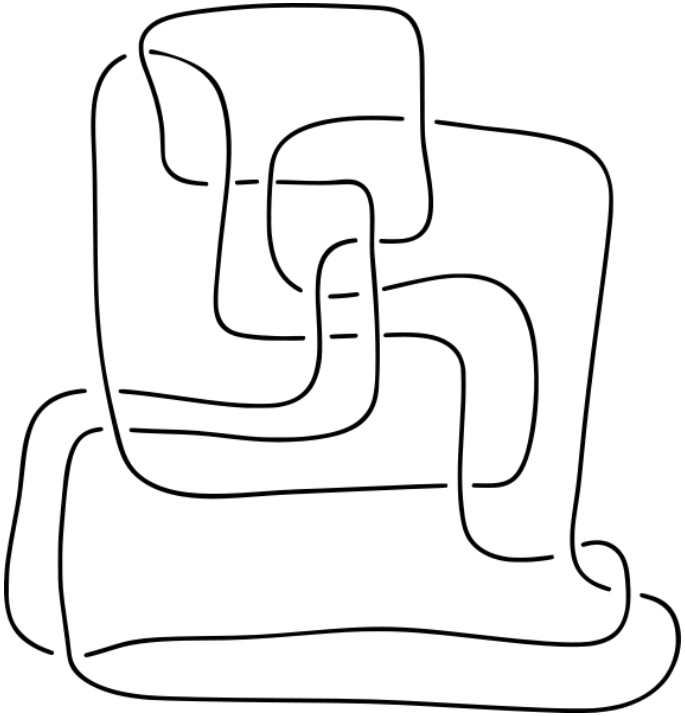
Projections of the unknot (circle) can be arbitrarily knotty-looking, as illustrated by the Thistlewaite unknot:



The fundamental question

To answer the fundamental question, we look at invariants of the knots in question: algebraic, geometric, or topological properties that hold for all manifestations of a given knot. The rationale being if two knots differ with respect to some invariant, they must be distinct.

Knot and Link Complements

One important invariant is the **knot complement**. Given a knot $K \hookrightarrow S^n$, the knot complement is $S^n - K$. Link complements are defined analogously.

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Theorem (Gordon-Luecke, 1989)

Two knots have the same type iff they have homeomorphic complements (this is conjecture 3.A.1 in Rolfsen).

The proof makes use of Dehn surgery which we'll see in Chapter 9.

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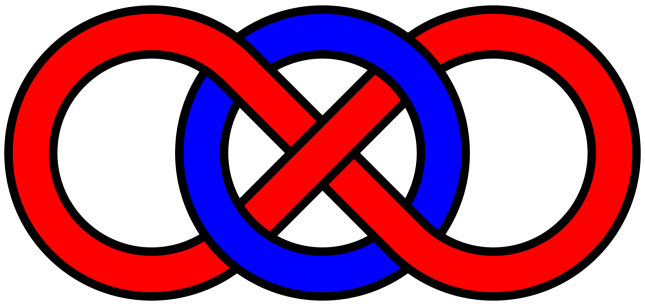
Theorem (Gordon-Luecke, 1989)

Two knots have the same type iff they have homeomorphic complements (this is conjecture 3.A.1 in Rolfsen).

We can't prove this yet, but we should try to later!

This does not hold for links! JHC Whitehead proved that there are infinitely many links whose complement is homeomorphic to that of the Whitehead link, pictured below.



Note on (co)homology of knot complement

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Alexander's duality theorem states that given a compact, locally contractible space $X \subset S^n$, we have

$$\widetilde{H}_q(S^n - X) \cong \widetilde{H}^{n-q-1}(X).$$

Then given a knot $K = Im(f : S^k \hookrightarrow S^n)$, we have

$$\widetilde{H}_q(S^n - K) \underset{\text{Alexander}}{\cong} \widetilde{H}^{n-q-1}(K) \cong \widetilde{H}^{n-q-1}(S^k) = \begin{cases} \mathbb{Z} & q = n-k-1 \\ 0 & \text{otherwise} \end{cases}$$

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Definition (Knot Group)

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The slogan for the argument is basically that when the codimension exceeds two, we have room to push our loops somewhere where we can safely contract them.

More detailed argument

The argument is that since we can embed each component of a link L^{n-3} in S^{n-1}. Given some component K of L^{n-3}, there's an ambient isotopy of $\mathbb{R}^n$ such that K resides in $\mathbb{R}^{n-1} \times \{0\} \subset \mathbb{R}^n$. Given any loop γ, we can nudge the portions of $Im(\gamma)$ residing in $\mathbb{R}^{n-1}$ along the nth dimensional axis until the entire image is outside of $\mathbb{R}^{n-1}$ and thus, homotopic to a constant loop. When the dimension of our link is less than $n-3$, we simply have more space in which we can rotate to avoid the embedded sphere in ambient n-space.

The Wirtinger Presentation

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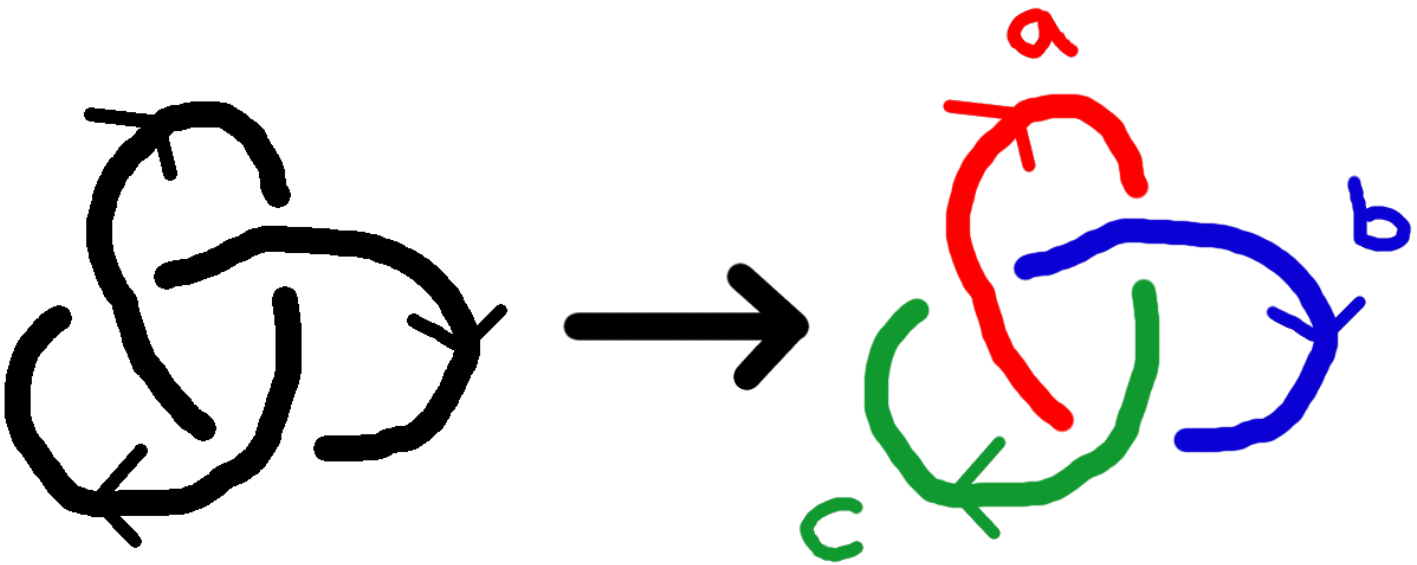
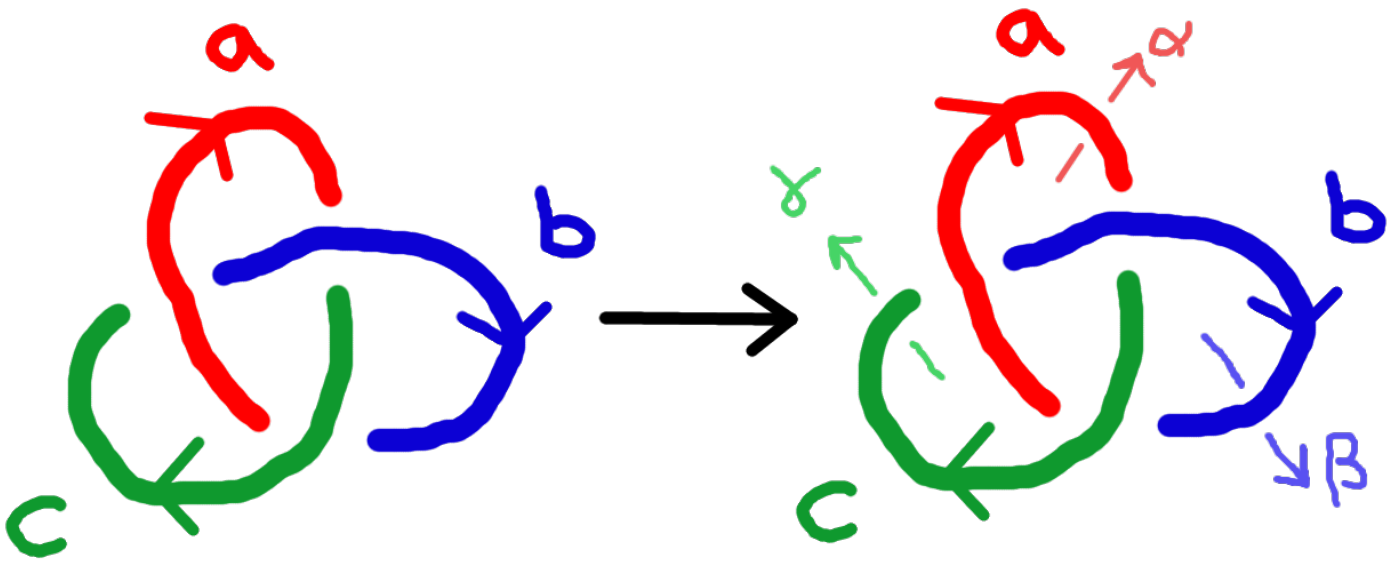
1. Take your knot projection, choose an orientation, and label the different connected components



The Wirtinger Presentation

2. Draw an arrow underneath each connected component such that if one were walking along the knot in the direction of the orientation, all arrows would have the same handedness. Below, all arrows go from "right" to "left." Each arrow represents the homotopy class of loops wrapping around the corresponding region of K.



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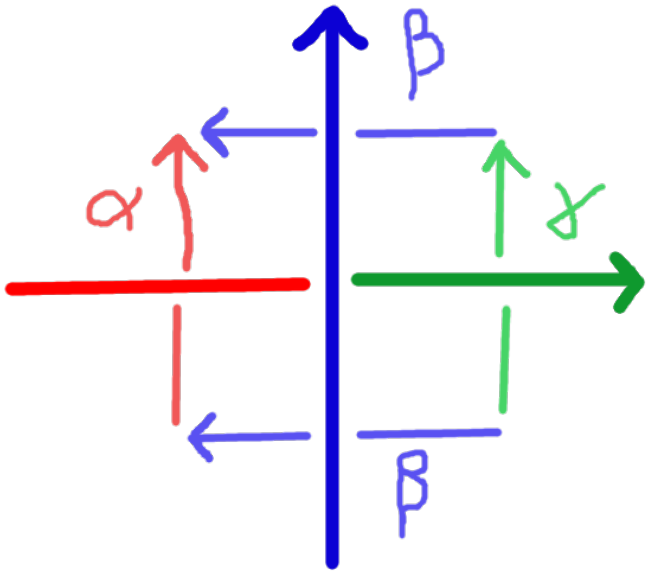
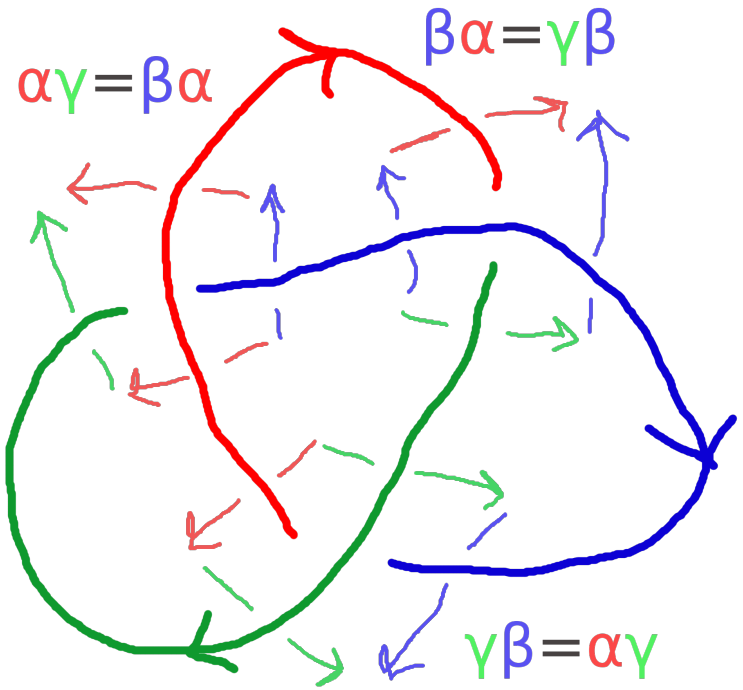
3. Look at the relations that our crossings impose on the generators of the fundamental group. For example, if we begin at component a, the first crossing (pictured below) imposes the condition $\beta\alpha = \gamma\beta$.



The Wirtinger Presentation

4. Iterating this process yields the relations
   - $\beta\alpha = \gamma\beta$
   - $\gamma\beta = \alpha\gamma$
   - $\alpha\gamma = \beta\alpha$ (this one is redundant)



The Wirtinger Presentation

5. (Optional) You do some algebra to reduce the presentation. We start with this

$$\langle \alpha, \beta, \gamma \mid \beta\alpha = \gamma\beta, \gamma\beta = \alpha\gamma, \alpha\gamma = \beta\alpha \rangle$$

but upon noticing that $\gamma = \beta\alpha\beta^{-1}$, we can reduce the presentation to

$$\langle \alpha, \beta \mid \beta\alpha\beta^{-1} = \alpha^{-1}\beta\alpha \rangle$$

Link Group

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Answer Find the Wirtinger presentation for each component in the link (i.e. find the group of each knot), then find the relations introduced by the crossings of the connected components.

Proving Wirtinger's algorithm yields the knot group

Suppose K is embedded in space such that most of it resides in the plane $z = 0$ with n dips of distance ϵ below said plane to create undercrossings.

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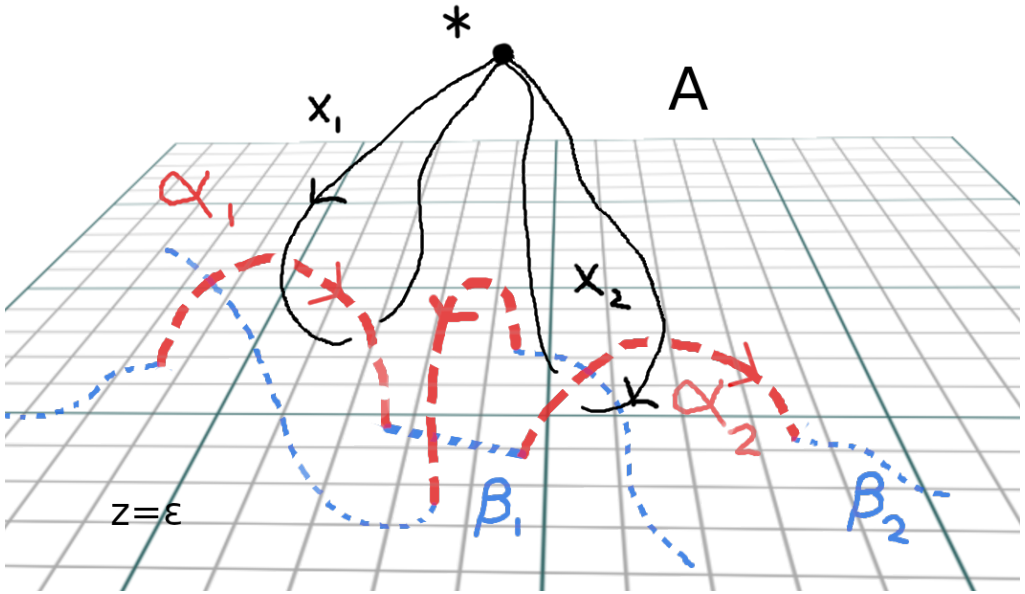
We'll let A be the region satisfying $-\epsilon \leq z$ minus K. We first observe that the fundamental group of A is generated by the x_i, each hooking "under" the bridges α_i.



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For each β_i we'll encase it in a box B_i such that the top of the box resides in $z = -\epsilon$. We remove β_i and also attach an arc from the box to the basepoint for our loops in A. We can go ahead and say that the B_i are disjoint.

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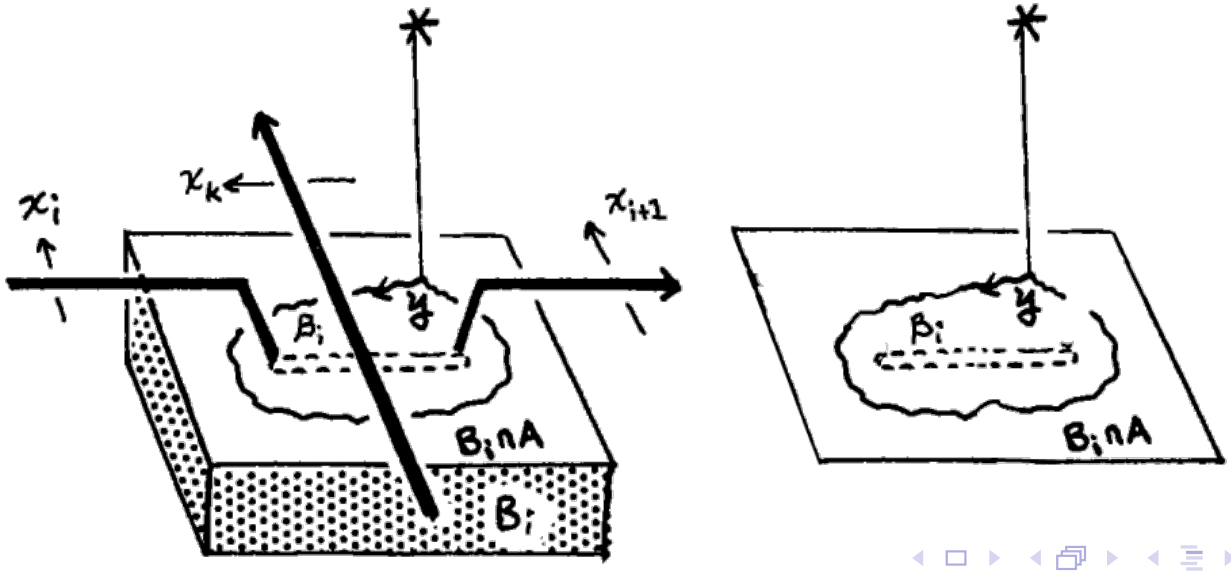

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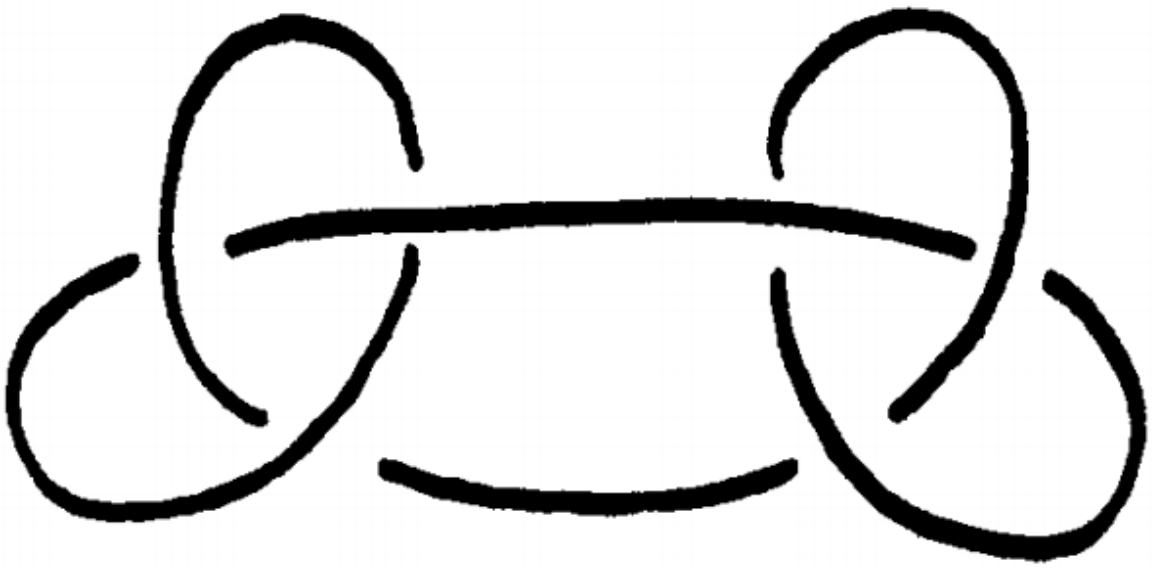
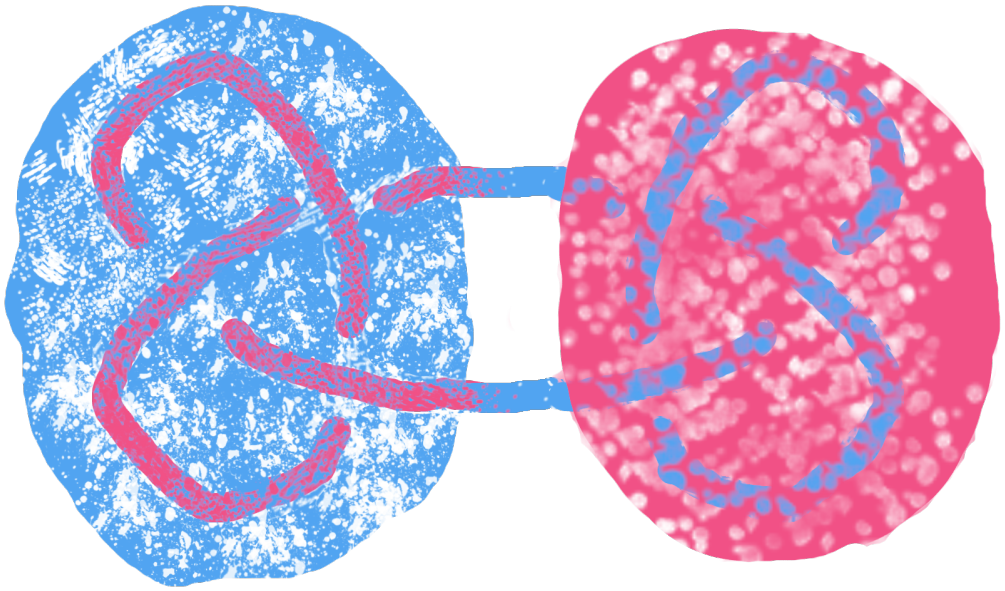
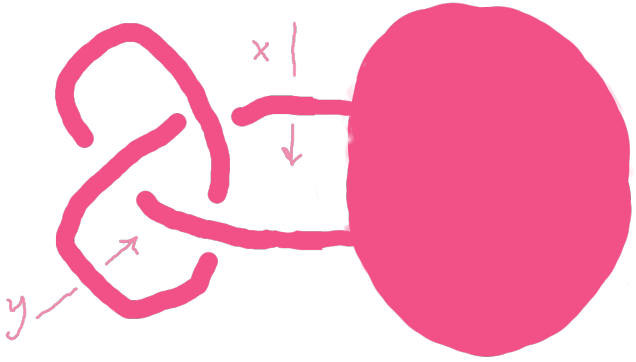
Example: The Square Knot This is the connected sum of two trefoil knots of opposite chirality (i.e. mirror images)..

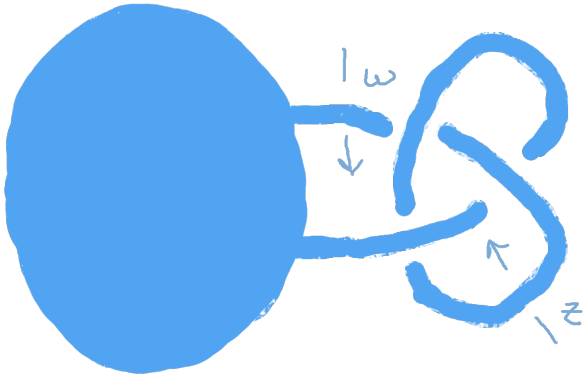


Composite Knots

Deforming our trefoils so that a point on the outside edge of each is replaced with a ball allows Van Kampen







Composite Knots

Applying Van Kampen gives us

$$\langle x, y, z, w \mid xyx = yxy, zwz = wzw \rangle / \langle xw^{-1} \rangle$$

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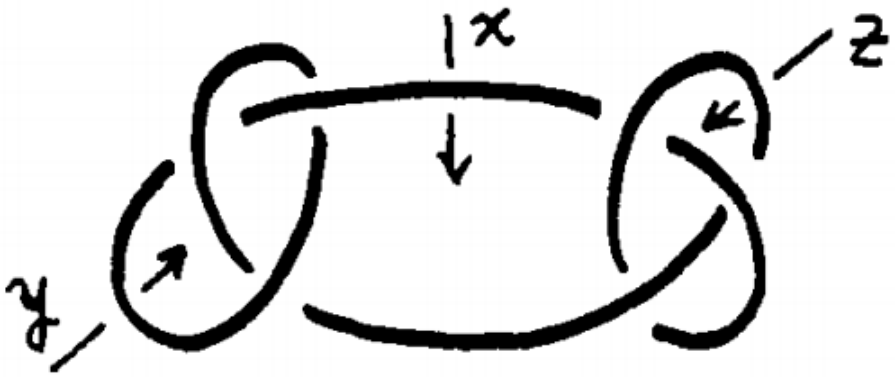
This group is the same as the group of the "Granny knot," a knot with nearly identical connected sum decomposition but with two equivalent copies of the trefoil.



Detour: Is the knot group actually special then?

Taking for granted the result that the trefoil knot is prime, one might notice that, if we ignore the chirality of the summands, the square and granny knot share an equivalent prime decomposition.

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It turns out, this holds in general as a consequence of Gordon-Luecke theorem! That is, two knots have isomorphic knot groups if and only if they have the same prime decomposition up to chirality of summands. I would be very interested in exploring this later.

Distinct links, homeomorphic complements

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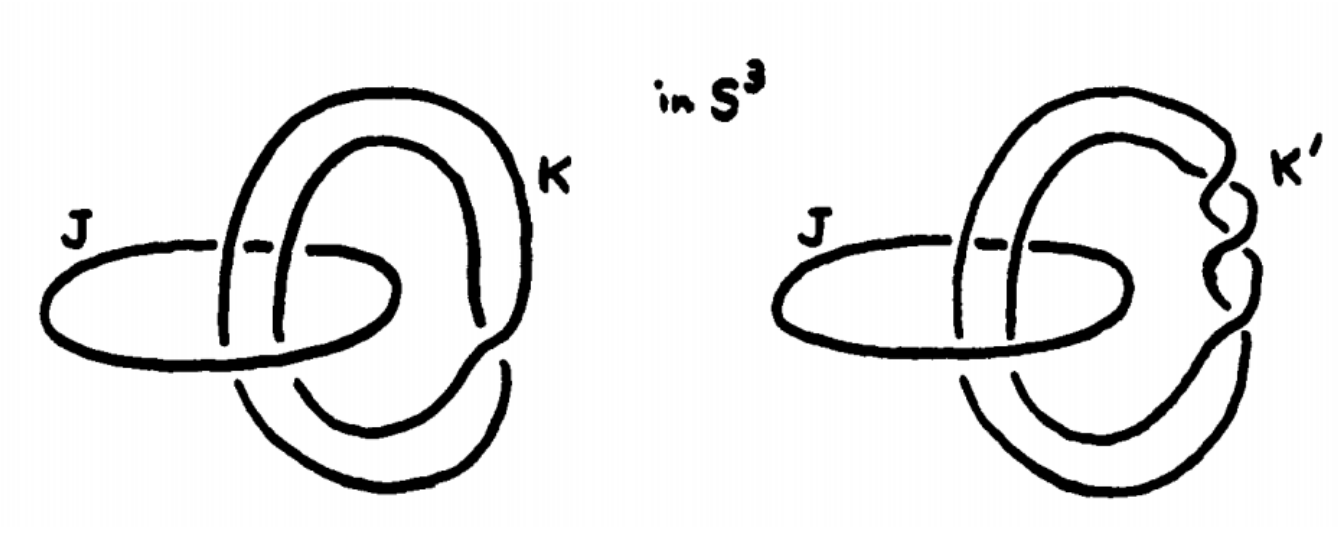
At this point, we know that the unknot and the trefoil are distinct knots, since unknot has a free abelian knot group and the trefoil doesn't. Hence, the two links pictured are distinct.



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One way to see this (thanks Rebecca!) is to regard $S^3 \subset \mathbb{C}^2$ and then to look at the subset

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Thinking about our coordinates now in polar form, i.e. $z_1 = (r_1, \theta_1)$ and $z_2 = (r_2, \theta_2)$, the equation together with the requirment for inclusion in S^3, i.e. $|z_1|^2 + |z_2|^2 = 1$ imposes the condition $r_i = \frac{1}{\sqrt{2}}$ while leaving the angular coordinates untethered. We get a circle of radius $\frac{1}{\sqrt{2}}$ in one complex plane and another copy in the "orthogonal" complex plane: $S^1 \times S^1$.

Distinct links, homeomorphic complements

Intuitively, we can decompose S^3 as two solid tori, one filling in the hole in the other and vice versa.

More precisely, we notice that

$$\mathbb{T} = \{(z_1, z_2) \in S^3 \mid |z_1| \leq |z_2|\}$$

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Since every point $(z_1, z_2) \in S^3$ will satisfy one of the conditions $|z_1| \leq |z_2|$ or $|z_2| \leq |z_1|$, we know that the union of the two solid tori is all of S^3 and that the points shared between these two sets is precisely the boundary of each: the values where $|z_1| = |z_2|$.

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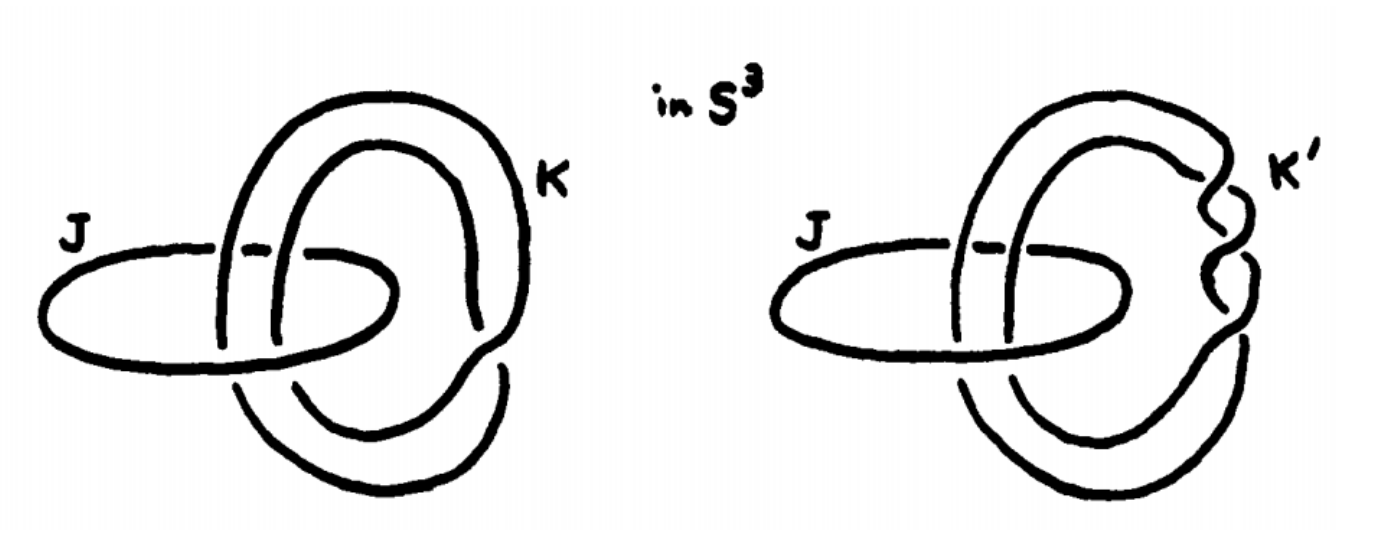

We now think of J as the result of contracting the interior of solid torus, meaning that K sits inside of the complementary solid torus. This torus is homeomorphic to one that has been twisted, yielding K'. Hence, there are two links with homeomorphic complements.

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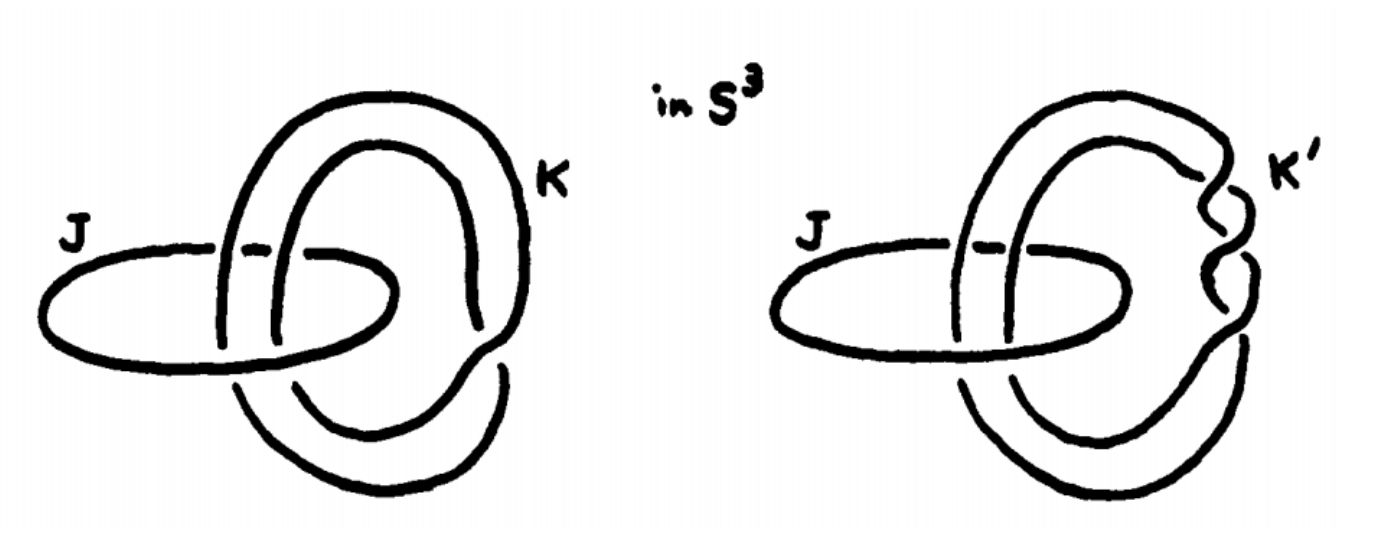

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We can keep twisting K' to produce new knots, yielding an infinite collection of distinct links with homeomorphic complements.

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Such 2D drawings are the images of **regular projections** $p : K \to P$ sending our knot K onto a plane P in a way that's "nice" in the sense that it does not send more than two points of K to the same point in P. Working with polygonal knots, we'll also assume that vertices share their image under p with no other points.

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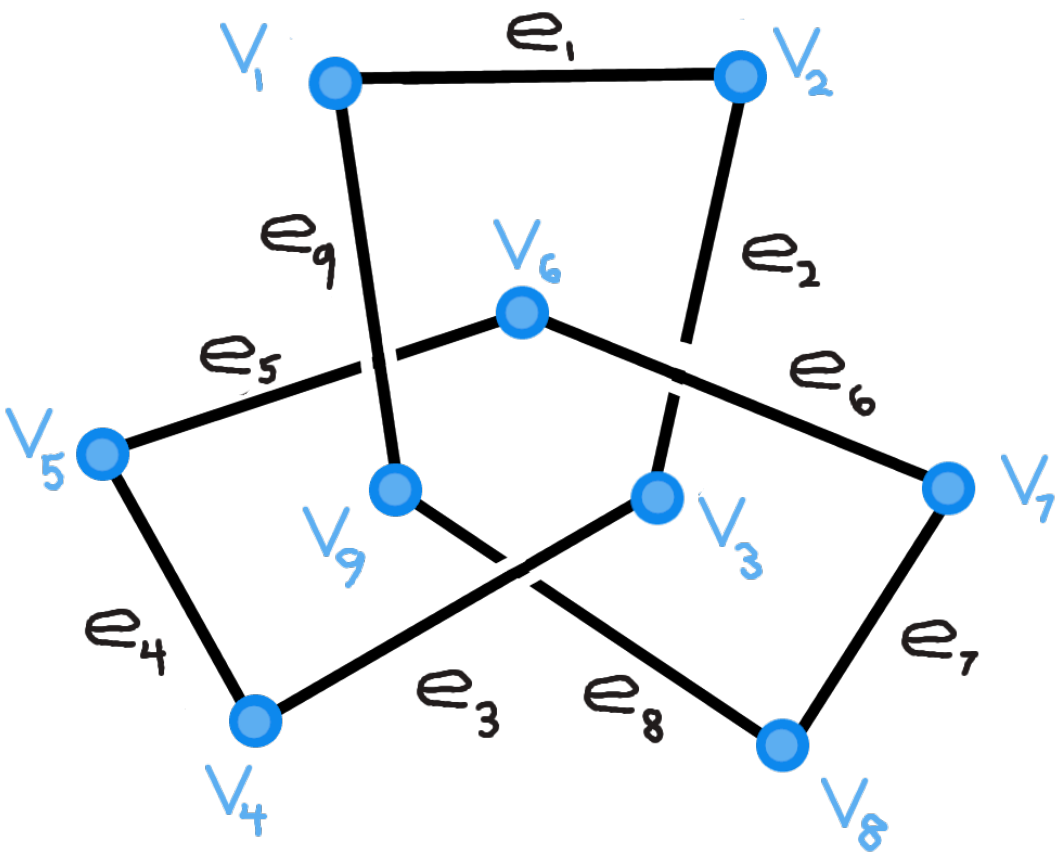
The answer is "no," but proving it takes a bit of work.

Regular projections exist!

Suppose we have a plane P and a polygonal knot $K \subset S^3$ with vertices $v_1, \ldots, v_n$ connected by edges $e_1, \ldots, e_n$ where the ordering convention is illustrated below:



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With our manipulation of P, our goal is to avoid the possibilities:

1. An edge is collapsed to a single point by p
2. $|p(e_i) \cap p(e_j)| > 1$ where e_i and e_j are distinct.

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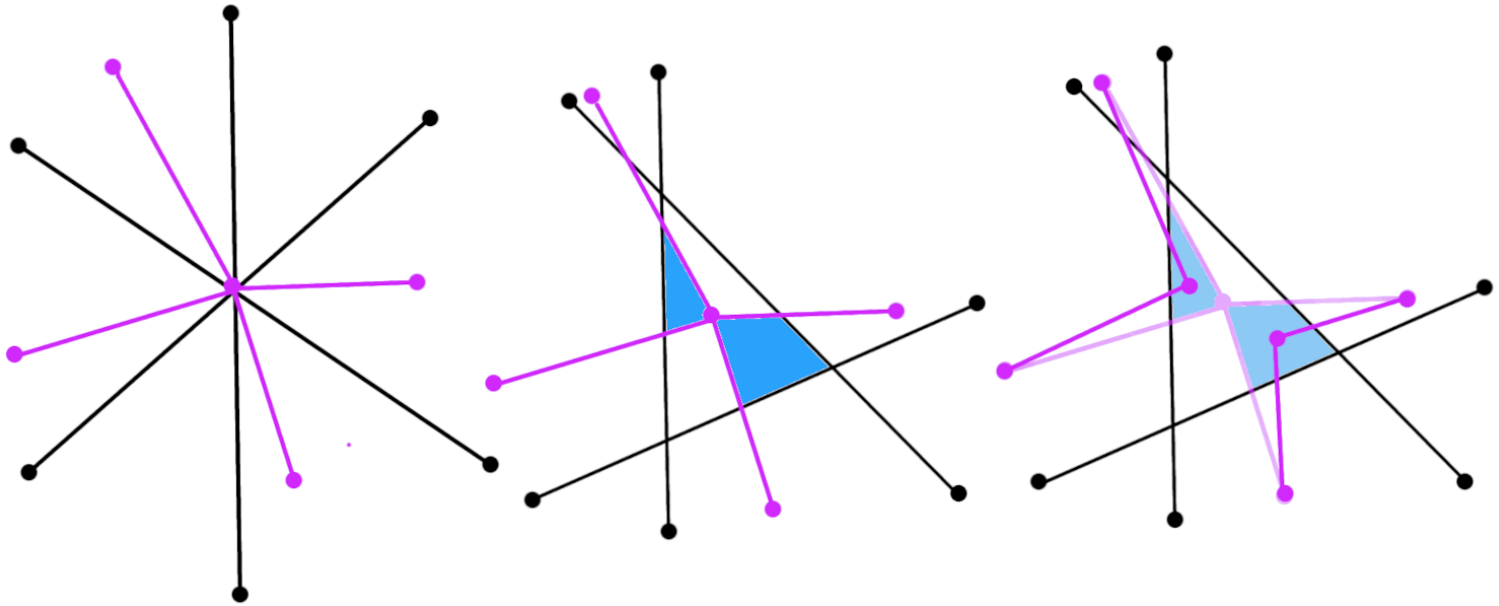
Case 1 only occurs when an edge e_i is parallel to the normal vector of P. There are finitely many e_i, and so only finitely many normal vectors to avoid. Case 2 occurs if there are coplanar edges $e_{i_1}, \ldots, e_{i_k}$ such that the normal vector is parallel to P. There are at most finitely many such planes and so the total set of "bad" choices for a normal vector to P is finite. Assuming P isn't already acceptable, we find the closest bad vector to the current vector normal to P and rotate in that direction by less than said minimal distance.

Regular projections exist!

Having suitably manipulated P, let $X_K = \{x \in P \mid 3 \geq |p^{-1}(x)|\}$. Given some $x_0 \in X_K$, we'll suppose for the moment that there are no vertices of K in $p^{-1}(x_0)$. We'll say that the distance between two edges e_i and e_j is the minimal distance between their respective points. Taking the minimum across all edge pair distances yields a value δ_1. We set $\delta_2 = min\{d(x_0, x) \mid x \in X_K\}$ and let $\delta = min\{\delta_1, \delta_2\}$. Transporting all edges containing a preimage of x_0 a distance less than $\frac{\delta}{2}$ along a vector parallel to P will adjust K such that x_0 is removed from X_K.

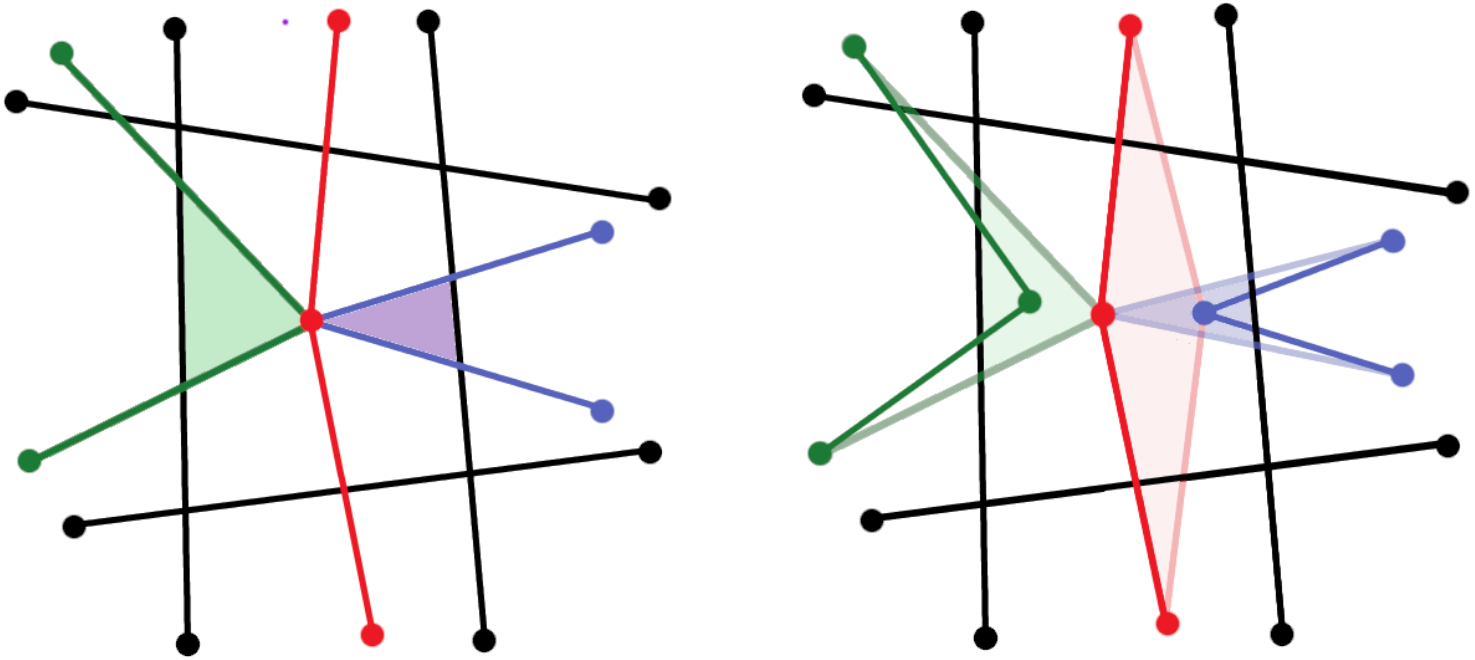
Regular projections exist!

If there are vertices in $p^{-1}(x_0)$, then we work through the process described above so that there are no non-vertex points in $p^{-1}(x_0)$. The projection of K into P is now such that x_0 is inside of a convex polygon C. Each vertex $v_i \in p^{-1}(x_0)$, consider the polygon bounded by the projections of all relevant edges. Transport each v_i along the plane spanned by e_i and e_{i+1} such that it's projection is contained in the bound region, as pictured below.



Regular projections exist!

In the situation where you have bound regions contained in others, we slightly modify the procedure. Transport vertices one at a time starting with vertex interior to the smallest angle. Take the convex hull of this vertex what of the next largest angle and the intersections as the side. Repeat until you have only one vertex in $p^{-1}(x_0)$.



Wirtinger Presentation is valid for all tame knots

For the second exercise, consider a polygonal knot K with vertices $\{v_i\}_{i\in[n]}$ and edges $\{e_i\}_{i\in[n]}$. To each e_i, we assign the value

$$\epsilon_i = inf\{d(x,y) \mid x \in e_i, y \in e_j, j \in [n] - \{i-1, i, i+1\}\}$$

Now setting $\epsilon = min\{\epsilon_i\}_{i\in[n]}$, we thicken K so that it is a solid tori with internal radius less than $\frac{\epsilon}{2}$ and note that the embedded torus has no self-intersections by construction. It follows then that given a new knot K' with vertex set $\{v_i'\}_{i\in[n]}$ satisfying $d(v_i, v_i') < \frac{\epsilon}{2}$, this polygonal knot is contained in the aforementioned embedded torus. Since none of the edges or vertices are allowed to travel far enough to self-intersect another portion of K, we are assured that the linear homotopy defined by $v_i \mapsto v_i'$ with edges following endpoints is in fact an ambient isotopy.

Wirtinger Presentation is valid for all tame knots

This last bit tells us that the setup from the proof of Wirtinger's algorithm is on solid ground for all the knots we're interested in!

Homotopically linked knots

TBD